**1** **(a)** 21 **[B2]**

M1 for 13 or 8

**(b)** 80 **[B2]**

M1 for 16 or 5

*[Total Marks: 4]*

**2** **(a)** 14 **[B1]**

**(b)** 59 **[B2]**

If 0 scored, M1 for 64 or 5 seen **[M1]**

**(c)** 5 **[B2]**

If 0 scored, M1 for $169 - 144$ or 25 seen **[M1]**

*[Total Marks: 5]*

**3** **(a)** $3x + 4 = 64$ **[M1]**

$3x = 60$ **[M1]**

20 **[A1]**

**(b)** $x^2 = 64 + 36 = 100$ **[M1]**

$10, -10$ **[A1]**

**(c)** $3\sqrt{x} = 21$ or $\sqrt{x} = 7$ **[M1]**

49 **[A1]**

*[Total Marks: 7]*

**4** **(a)** 125 **[B2]**

If 0 scored, M1 for evaluating $16 - 9 - 2$ to get 5 **[M1]**

**(b)** $-2$ **[B3]**

M1 for $\sqrt{64}$ or 8 **[M1]**

M1 for $5 - 9$ or $-4$ **[M1]**

**(c)** 729 **[B3]**

M1 for $2\sqrt[3]{y} = 6$ or $\sqrt[3]{y} = 3$ **[M1]**

M1 for $y = 27$ **[M1]**

*[Total Marks: 8]*

**5** **(a)** 5.3990299... **[B2]**

M1 for numerator evaluated to 163.91... or denominator evaluated to 30.359...

**(b)** 5.399 **[B1]**

**(c)** 6 **[B2]**

M1 for evaluating $6^4 = 1296$ and $7^4 = 2401$

*[Total Marks: 5]*

**6 (a) (i)** 1.49315... **[B2]**

If 0 scored, M1 for 3.329 seen or 72.25 and 68.921 **[M1]**

**(ii)** 1.49 **[B1]**

**(b)** 96 **[B2]**

If 0 scored, M1 for 23.04 or 0.24 seen **[M1]**

*[Total Marks: 5]*

**7 (a)** Calculates $3^3 = 27$ and $\sqrt{144} = 12$ **[M1]**

15 **[A1]**

**(b)** Calculates $2^4 = 16$ and $\sqrt[3]{125} = 5$ **[M1]**

80 **[A1]**

*[Total Marks: 4]*

**8 (a)** Numerator correctly evaluated to $64 + 15 = 79$ **[B1]**

Denominator correctly evaluated to 4 **[B1]**

19.75 **[B1]**

**(b)** Substitution leading to $25 - (-8)$ **[M1]**

33 **[A1]**

*[Total Marks: 5]*

**9 (a)** 1.5 **[B1]**

**(b)** Rearranges to $2x^4 = 1250$ leading to $x^4 = 625$ **[M1]**

$x = 5$ **[B1]**

$x = -5$ **[B1]**

**(c)** Correct substitution: $\sqrt{\dfrac{3(4.2)^2 + (-5.7)}{1.8}}$ **[M1]**

Evaluates numerator to 47.22 OR fraction to 26.2333... **[M1]**

5.12 **[A1]**

*[Total Marks: 7]*

**10 (a)** 14 **[B1]**

**(b)** 48 **[B2]**

If 0 scored, allow [M1] for correctly evaluating either $\sqrt[3]{27} = 3$ or $2^4 = 16$

**(c)** 8.8 or $\frac{44}{5}$ **[B2]**

If 0 scored, allow [M1] for a correct numerator of 44 or a correct denominator of 5

*[Total Marks: 5]*

**11 (a)** 5 **[B1]**

**(b)** 150 **[B2]**

If 0 scored, allow [M1] for $6 \times 5^2$ or $6 \times 25$

**(c)** 121 **[B2]**

If 0 scored, allow [M1] for evaluating $3^3 - 4^2 = 11$

*[Total Marks: 5]*

**12 (a)** 0.239 **[B2]**

If 0 scored, allow [M1] for evaluating the numerator as 19.679... or the denominator as 82.464...

**(b)** 64 **[B3]**

If 0 scored, allow [M1] for isolating $x^2$ correctly, e.g., $x^2 = 16^3$ or $x^2 = 4096$

Allow a further [M1] for $x = \sqrt{4096}$

**(c)** 7.44 or 7.443... **[B3]**

If 0 scored, allow [M1] for finding the volume of the cube as $12^3$ or 1728

Allow a further [M1] for correctly rearranging the sphere volume formula to solve for $r$, e.g., $r^3 = \frac{1728 \times 3}{4\pi}$ or $r = \sqrt[3]{\frac{1296}{\pi}}$

*[Total Marks: 8]*